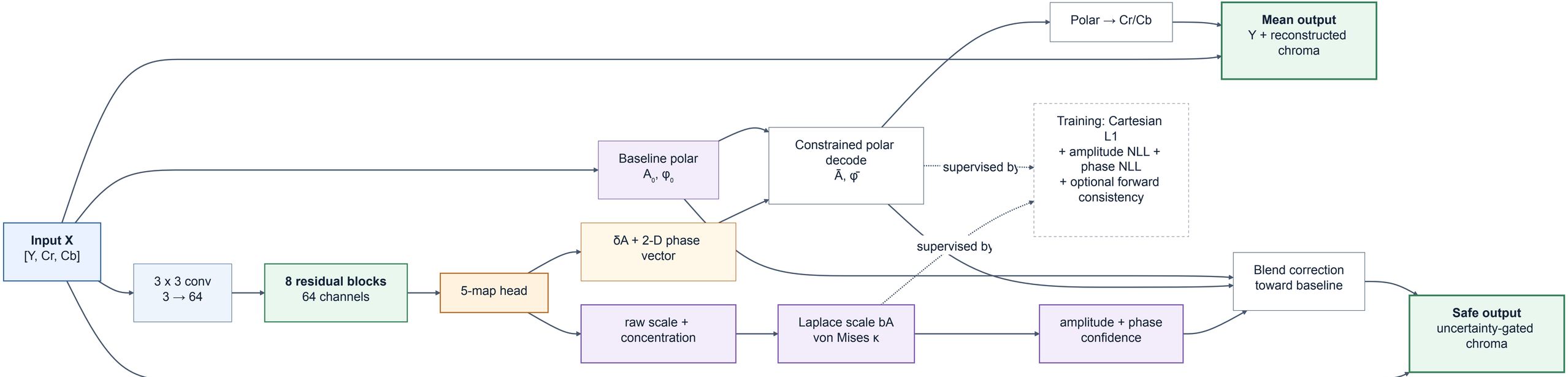

Input X
[Y, Cr, Cb]
3 x 3 conv
3 → 64
8 residual blocks
64 channels
5-map head
Baseline polar
A0, φ0
δA + 2-D phase
vector
raw scale +
concentration
Constrained polar
decode
Ā, φ̄
Laplace scale bA
von Mises κ
Polar → Cr/Cb
Mean output
Y + reconstructed
chroma
supervised by
Training: Cartesian
L1
+ amplitude NLL +
phase NLL
+ optional forward
consistency
supervised by
amplitude + phase
confidence
Blend correction
toward baseline
Safe output
uncertainty-gated
chroma